Theory of the Firm: Perfect Competition and Monopoly

Market Structures

A **market** is any physical or virtual arrangement where buyers and sellers interact to exchange goods or services.

The structure of market depends on three things: i.the number of firms, ii.how similar the products are, and iii.how easily new firms can enter. Based on these three criteria, market structures are divided into 4 categories. Such as

1. Perfect competition

2. Monopolistic competition

3. Oligopoly

4. Monopoly

Characteristic	Perfect competition	Monopolistic competition	Oligopoly	Monopoly
Number of firms	Many	Many	Few	One
Type of product	Identical	Differentiated	Identical or differentiated	Unique
Ease of entry	High	High	Low	Entry blocked
Examples	Apples, wheat and other agricultural products	Restaurants, DVDs, cloths, shoes, cosmetics	Computers, cars	Tap water (RWASA), letter delivery

Perfect Competition

Features of Perfect Competition

Perfect competition is said to prevail when the following conditions are found in the market

1. Many buyers and sellers, all small relative to the market.
2. All firms sell identical (homogeneous) products. It means that the **products of various firms are indistinguishable from each other**; they perfectly substitute for one another.
3. No barriers to entry or exit of firms.
4. Buyers and sellers have perfect information about the prevailing price.
5. **Price taker:** a buyer or seller that cannot affect the market price. The market sets the price (demand and supply intersect). Each firm faces a **horizontal, perfectly elastic demand curve** at that price.

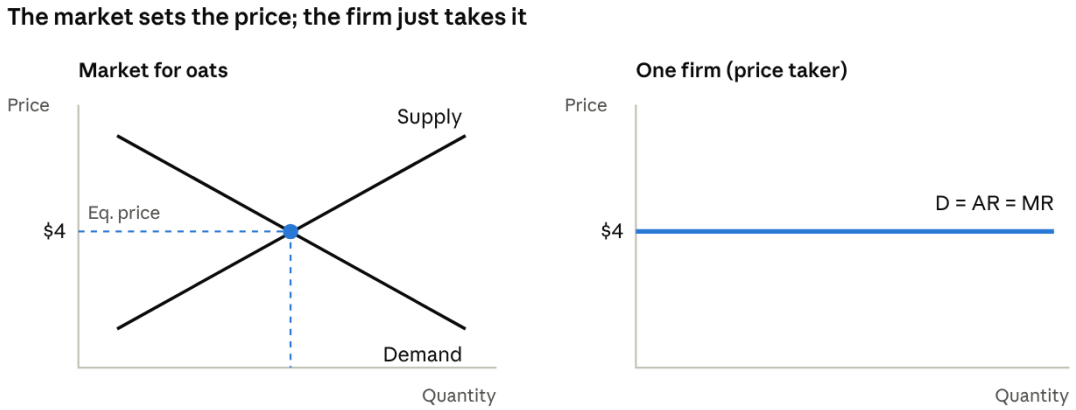


Figure 1: market vs firm demand in a perfect competition

Market demand and supply set the price; the individual firm's demand is a horizontal line. Market demand and supply fix the price at $4; the single firm sees a flat demand line at the price $4.

Revenue: TR, AR, MR

Total revenue: Total Revenue is the total money a producer earns from selling goods or services, before subtracting any costs.

$$TR = P \times Q$$

Average revenue: Average Revenue is the revenue earned per unit of output sold.

$$AR = TR / Q = (P \times Q) / Q = P$$

Marginal revenue: Marginal Revenue is the additional revenue earned from selling one more unit of a product.

$$MR = \Delta TR / \Delta Q$$

Revenue of firm under perfect competition

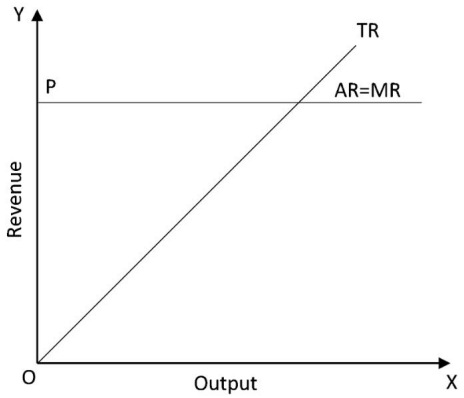
Under perfect competition, price is constant for every unit sold, so **P = AR = MR**.

- The TR curve is a straight line through the origin, rising in proportion to sales.
- The slope of TR is MR, which is constant and equal to the market price.

Example: Revenues from sells of an oats farmer

Q (KG)	Price (P)	TR	AR	MR
1	4	4	4	4
2	4	8	4	4
3	4	12	4	4

Q (KG)	Price (P)	TR	AR	MR
4	4	16	4	4
5	4	20	4	4



Firm's Equilibrium and Profit Maximisation

A firm is in equilibrium when it has no incentive to change its output, which means it is at its profit-maximising output. In other words, a state of production where, once the firm reaches it, it has **no tendency to move to any other output**. If the firm concludes that its profit is at the maximum here and cannot be raised at any other output, that output point is the **equilibrium point**.

The Total approach (TR and TC)

Firm’s equilibrium and profit is maximised where the difference between TR and TC is highest.

Profit= **TR - TC**

The Marginal approach (MR and MC)

Firm’s equilibrium and profit is maximised where **MR = MC**.

Conditions of equilibrium

1. **Necessary (first-order) condition:** at the equilibrium point, marginal revenue equals marginal cost, that is **MR = MC**.
2. **Sufficient (second-order) condition:** at the equilibrium point, the MC curve must **cut the MR curve from below and go upward**, that is **slope of MC > slope of MR**.

Even when both conditions hold, the firm must still compare **average revenue (AR) with average cost (AC)** to see whether it earns abnormal profit, normal profit or a loss:

Comparison at equilibrium output	Result in the short run
AR > AC	Equilibrium with **abnormal (supernormal) profit**
AR = AC	Equilibrium with **normal profit**
AR < AC	Equilibrium with a **loss** (the firm bears the loss and still produces)

Equilibrium of a Firm under Perfect Competition

a perfectly competitive firm can earn abnormal profit, normal profit or a loss in the short run. The firm reaches its maximum profit (or minimum loss) at equilibrium, and the price is settled through that equilibrium.

(a) Equilibrium with abnormal profit

If the firm keeps its average cost (AC) below its average revenue (AR) through efficient production, it earns abnormal profit. In the figure, MC cuts MR from below at **E**, so both conditions are met. The equilibrium price is **OP** and the equilibrium output is **OM**.

- At output OM, AR (= EM) is greater than AC (= FM).
- Abnormal profit per unit = EM − FM = **EF**.
- Total abnormal profit = OM × EF = GF × EF = **area PEFG**.

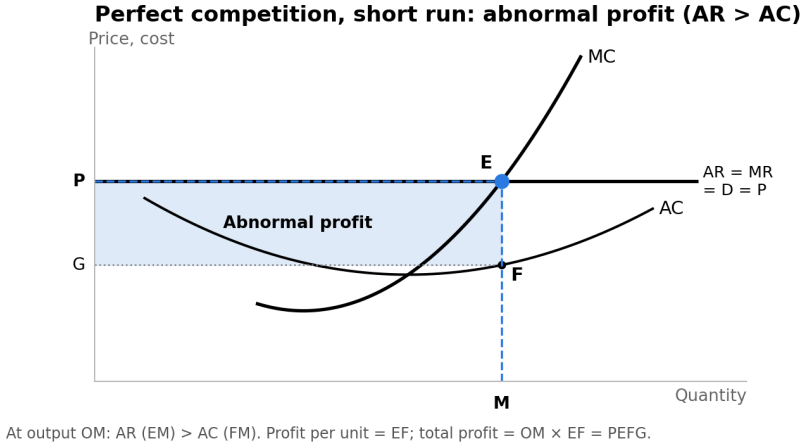


Figure 13: short-run equilibrium of a competitive firm with abnormal profit (shaded rectangle PEFG).

(b) Equilibrium with normal profit

If the firm produces so that AR equals AC, it earns only normal profit. At **E** both conditions of equilibrium are met, so price **OP** and output **OM** are fixed. At output OM, AR = AC = EM. Revenue just covers all costs (including the normal return to the entrepreneur), so there is **no extra profit**.

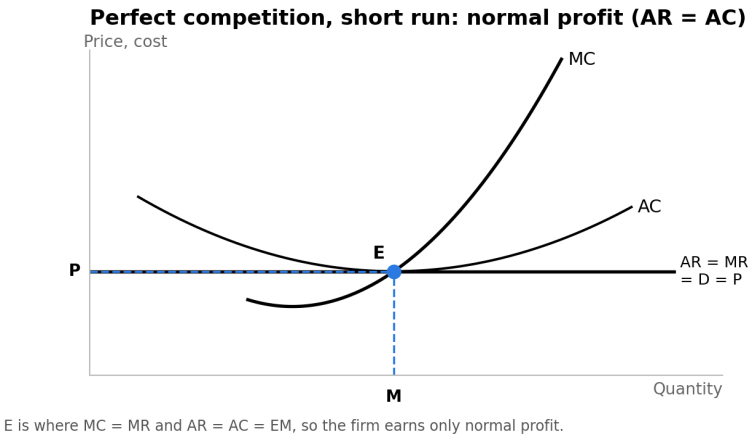


Figure 14: short-run equilibrium of a competitive firm with normal profit (AC touches the price line at E).

(c) Equilibrium while bearing a loss

If AR is less than AC, the firm faces a loss in the short run. At **E** both conditions of equilibrium are met, giving price **OP** and output **OM**. At OM the firm's AC (= FM) is greater than its AR (= EM).

- Loss per unit = FM − EM = **EF**.
- Total loss = EF × OM = EF × GF = **area PEFG**.

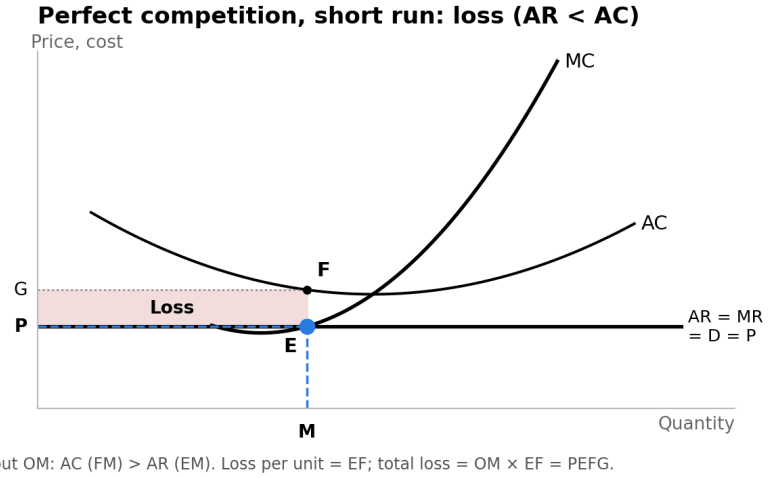


Figure 15: short-run equilibrium of a competitive firm bearing a loss (shaded rectangle PEFG).

Monopoly Market

Monopoly: a firm that is the only seller of a good or service that has no close substitute.

Key Features of a Monopoly Market

- **Single Seller:** Only one firm produces and sells the products.

- **No Close Substitutes:** The unique product sold by the monopolist has no direct or close alternatives.
- **High Barriers to Entry:** Strict obstacles prevent other competitors from entering the market. These barriers can include legal patents, government licenses, high startup/infrastructure costs, or exclusive control over vital raw materials
- **Price Maker:** Because the firm controls the total supply, it has the power to set the price of the product rather than taking it from the market. It faces a **downward-sloping demand curve.**

Revenue of firm under Monopoly

AR and MR and the demand curve: A monopolist faces the whole market demand curve, which slopes downward. To sell more it must cut the price. **That is why a monopoly firm is called a Price Maker.** Cutting the price has two effects:

- *Good:* it sells more units.
- *Bad:* it receives less revenue on every unit than it would have at the higher price.

Therefore:

- **AR = P**, so the AR curve is the demand curve.
- **MR < P** for every unit after the first, so the **MR curve lies below the demand curve**.
- With a straight-line demand curve, MR has the same intercept and **twice the slope**. It cuts the quantity axis at half the demand intercept.

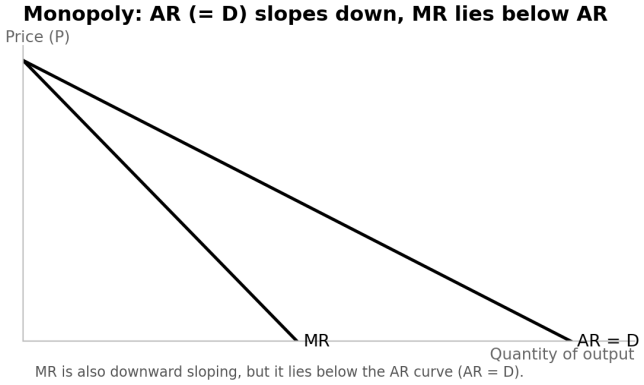


Figure 11: downward-sloping AR (demand) with MR below it

(AR and MR curves under monopoly): in a monopoly market the average revenue curve and the demand curve are the same (AR = D). Both slope downward. The MR curve also slopes downward, but it lies **below** the AR curve.

Equilibrium of a Firm under Monopoly

The monopolist is also a profit maximiser, so it also produces where **MR = MC**. The difference is how the price is found. So, for a monopolist **P > MR = MC**, whereas for a competitive firm P = MR = MC.

(a) Equilibrium with abnormal profit

If the monopolist can keep AC below AR by producing efficiently, it earns abnormal profit. In the figure, MC cuts MR from below at **E**, so the equilibrium is reached with price **OP** and output **OM**. At OM, AR (= RM) is greater than AC (= SM).

- Abnormal profit per unit = RM − SM = **RS**.
- Total abnormal profit = OM × RS = PR × RS = **area PRST**.

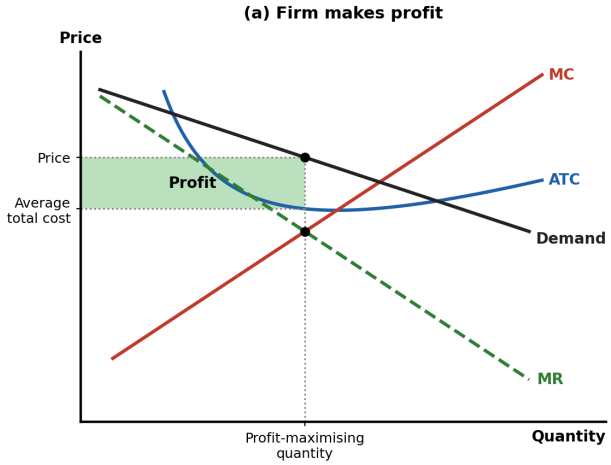


Figure 16: short-run monopoly equilibrium with abnormal profit (shaded rectangle PRST).

(b) Equilibrium with normal profit

If AR and AC are equal through production, the monopolist earns normal profit. At **E** the equilibrium conditions are met, so price is **OP** and output is **OM**. At OM, AR = AC, so **no additional profit** is earned. The monopolist reaches equilibrium with normal profit only.

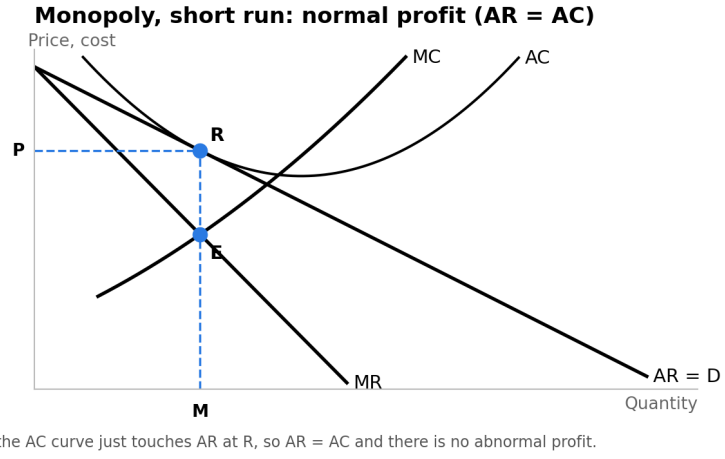


Figure 17: short-run monopoly equilibrium with normal profit (AC just touches AR).

(c) Equilibrium while bearing a loss

If AR is less than AC at the output level, the monopolist bears a loss yet still reaches equilibrium. At **E** the conditions are met, so price is **OP** and output is **OM**. At OM, AC (= SM) is greater than AR (= RM).

- Loss per unit = SM − RM = **SR**.
- Total loss = OM × SR = PR × SR = **area PRST**.

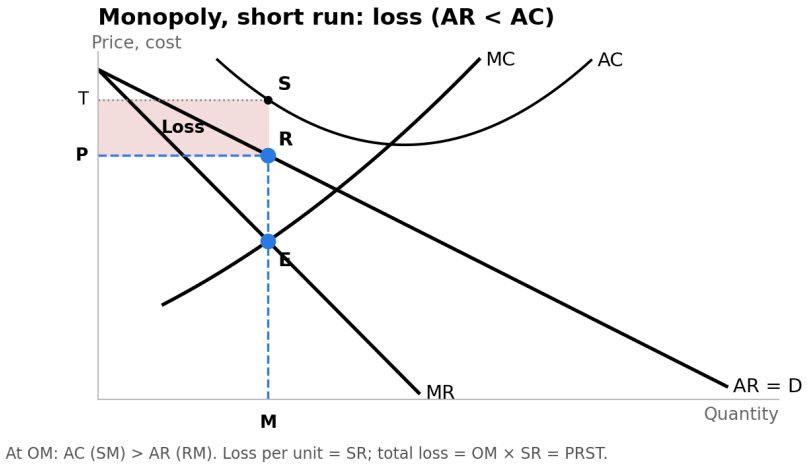


Figure 18: short-run monopoly equilibrium bearing a loss (shaded rectangle PRST).

Conclusion: in the short run, a monopoly firm can reach equilibrium with abnormal profit, normal profit or even a loss, depending on the situation. A monopolist is not guaranteed profit in the short run, but barriers to entry let it keep any abnormal profit in the long run.

Perfect Competition vs Monopoly

Feature	Perfect competition	Monopoly
Firms	Many	One
Demand faced	Horizontal	Downward sloping (market demand)
AR vs MR	AR = MR = P	AR = P > MR
Equilibrium rule	MR = MC (so P = MC)	MR = MC (P above MC)
Long-run profit	Zero economic profit	Can be positive
Supply curve	MC above AVC	None
Efficiency	Efficient	Deadweight loss